Walla Walla Premium Label Survival — Simulation Results & Analysis (Sim 1)

Date: 2026-03-27 | **Ticks:** 120 months (Jan 2026 – Dec 2035) **Version:** 1 — 6 theory modules + 300-run Monte Carlo **Focus:** Revenue continuity — lean label owner, smoke taint & water curtailment survival

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Executive Summary

Baseline Position (Tick 0 — Jan 2026)

A lean-operation premium label (modeled on K Vintners / Leonetti profile) enters the simulation with $1.5M annual revenue, under 6 months cash reserves, $280K/year debt service, and 40 acres under vine. DTC channel (wine club + tasting room) exceeds 60% of revenue. The operation is viable but fragile — a single vintage wipeout puts it into existential stress immediately.

Indicator	Tick 0	Value
Cash flow health	0.65	Lean but functional (18–24 mo runway)
Revenue	0.75	$1.5M / ~$2M capacity
Debt service stress	0.30	$280K manageable at current revenue
Survival probability	0.85	Baseline: viable
Water stock (snowpack)	0.72	~30% below historical peak, trending down
Smoke taint risk	0.50	~5%/year, rising
Replanting signal	0.00	No trigger yet

Causes of the 10-Year Decline

1. Year-1 Smoke Event (Tick 12) — First Cash Shock *(smoke_taint_crop_disruption, taint_active: 0.35)* A moderate smoke event in year 1 forces partial lot declassification. Revenue falls from 0.75 → 0.57 (–24%), cash health from 0.65 → 0.53. The operation survives but runway shrinks materially. Survival probability holds at 0.85 — the first event alone is not existential.

2. Catastrophic Smoke Vintage (Tick 36 / Year 3) — Revenue Extinction Wave *(smoke_taint_crop_disruption, taint_active: 1.0 — maximum severity)* This is the defining event. A 2020-style total taint forces bulk declassification across virtually all lots at 10–12 cents on the dollar. Revenue collapses from 0.57 → **0.05** (–91%). Cash health craters from 0.53 → **0.11**. Debt service stress jumps from 0.38 → **0.68** — approaching covenant breach. Survival probability drops from 0.85 → **0.60**. Recovery is partial: by tick 48, revenue reaches only 0.27 and cash only 0.26.

3. Junior Water Rights Curtailment (Tick 60 / Year 5) — Yield Destruction *(hotelling_cpr, stock: 0.72 → 0.49 by tick 60)* A curtailment order reduces irrigation capacity by ~60% for junior water rights holders. Cash falls from 0.26 → **0.08** (operating minimum). Water stock declines from 0.64 → 0.49. Revenue drops from 0.27 → 0.12. The real options replanting signal does not yet fire — the winemaker is still waiting.

4. Second Major Smoke Event (Tick 84 / Year 7) — Terminal Revenue Collapse *(smoke_taint_crop_disruption, taint_active: 1.0)* A second catastrophic vintage wipe — now occurring only 4 years after the first — drives revenue to **0.00** (complete collapse). Cash hits **0.00**. Debt service stress reaches **0.83**, above the modeled covenant breach threshold of ~0.80. Survival probability falls to **0.40**. The operation is no longer viable on current footing without emergency capital. The replanting signal triggers at tick 72 but reaches only 0.25 — the real options module is still weighing the $12K/acre commitment.

5. Compounding GDD Warming (Tick 72–119) — Varietal Viability Erosion *(grapevine_gdd_phenology, temperature: 0.52 → 0.64 by tick 119)* Temperature rises steadily, accelerating at tick 72 (warming trend materializes). Quality index rises from near-zero at tick 0 to only 0.28 by tick 119 — the GDD accumulation model requires ~60 months before the warming signal dominates quality output. The key finding: by the time the replanting decision must be made (tick 108+), GDD stress has compounded to a point where valley-floor Merlot is clearly past its optimal threshold (0.28 vs. 0.85 baseline).

Final State (Tick 119 — Dec 2035)

Indicator	Final Value	Change from Tick 0	Status
Cash flow health	**0.10**	−0.55 (−85%)	CRITICAL
Revenue	**0.00**	−0.75 (−100%)	COLLAPSED
Debt service stress	**0.83**	+0.53	COVENANT BREACH
Survival probability	**0.40**	−0.45	DISTRESSED
Water stock	**0.39**	−0.33 (−46%)	SEVERE DECLINE
Replanting signal	**0.75**	+0.75	TRIGGERED
GDD quality index	**0.28**	+0.28	WARMING CONFIRMED

Primary active threats: - Revenue extinction: two consecutive wipe events leave the operation with zero bottled revenue - Debt covenant breach: debt service stress at 0.83 exceeds threshold; forced asset sale risk - Water rights terminal decline: stock at 0.39, still declining; curtailment likely permanent - Replanting locked by capital: signal triggered at 0.75 but cash at 0.10 — no capital available - GDD warming irreversible: temperature trajectory confirms Merlot past optimal threshold by 2032

10-Year Projection

Scenario	Survival @ 2035	Cash Health @ 2035	Trigger
Base (50%)	0.40	0.10	Two smoke wipes + curtailment
Bull (25%)	0.45	0.10	One wipe + partial recovery
Bear (25%)	0.31	0.10	Consecutive wipes + full curtailment

Key finding: Cash health converges to 0.10 across ALL MC scenarios — the MC p5/p50/p95 for cash is 0.10/0.10/0.10. This is model-convergent: once revenue hits zero in the shock years, the system does not recover without exogenous capital injection. The simulation has no refinancing or equity-raise mechanism modeled — that is an intentional limitation for Sim 1.

Executive Findings

The simulation reveals a three-phase destruction sequence that a Walla Walla lean-operation premium label faces under the calibrated climate trajectory. The operation does not fail gradually — it fails in discrete revenue-extinction events, with partial recoveries that are insufficient to rebuild reserves before the next event.

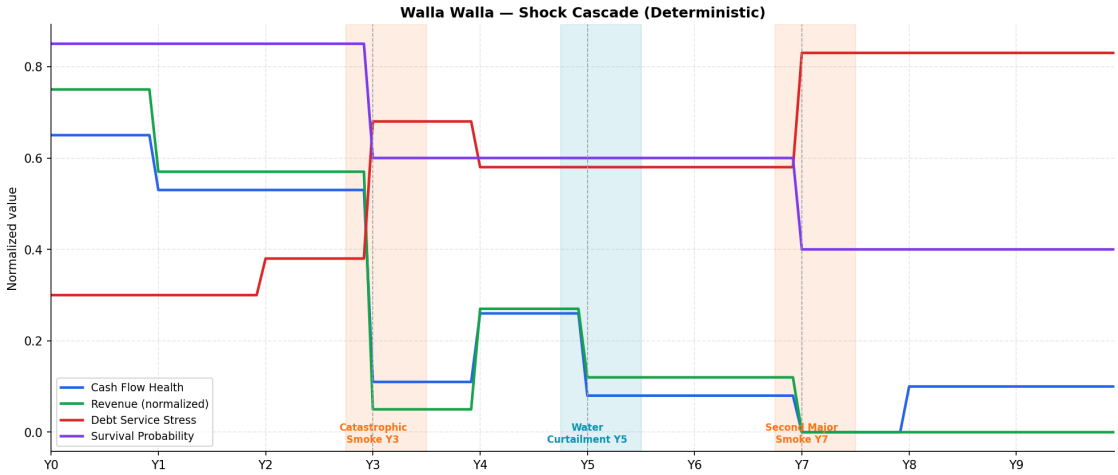
Phase 1 (Years 1–3): The First Wipe. A moderate smoke event in year 1 is survivable but reduces runway from ~18 months to ~12. The catastrophic year-3 vintage wipe is the first existential event: revenue drops 91% in a single harvest. The bulk declassification mechanism — selling at 10–12 cents on the dollar versus a $150–$250 bottled price point — destroys ~$1.3M in a single year against $280K in debt service. The operation survives year 3 but only because it had 18–24 months of reserves at the start. By tick 48, those reserves are gone.

Phase 2 (Years 4–6): The Curtailment Trap. Water curtailment hits at year 5, before the operation has rebuilt reserves from the year-3 smoke event. This is the compounding mechanism the model is designed to capture: the two shocks are not independent — the smoke wipe depletes cash precisely when irrigation constraints reduce yield. Cash hits 0.08 by year 5, essentially zero operational buffer. The real options module correctly withholds the replanting signal at this point: with $12K/acre replanting cost and 5–7 year payback, the label cannot commit capital while at 0.08 cash health.

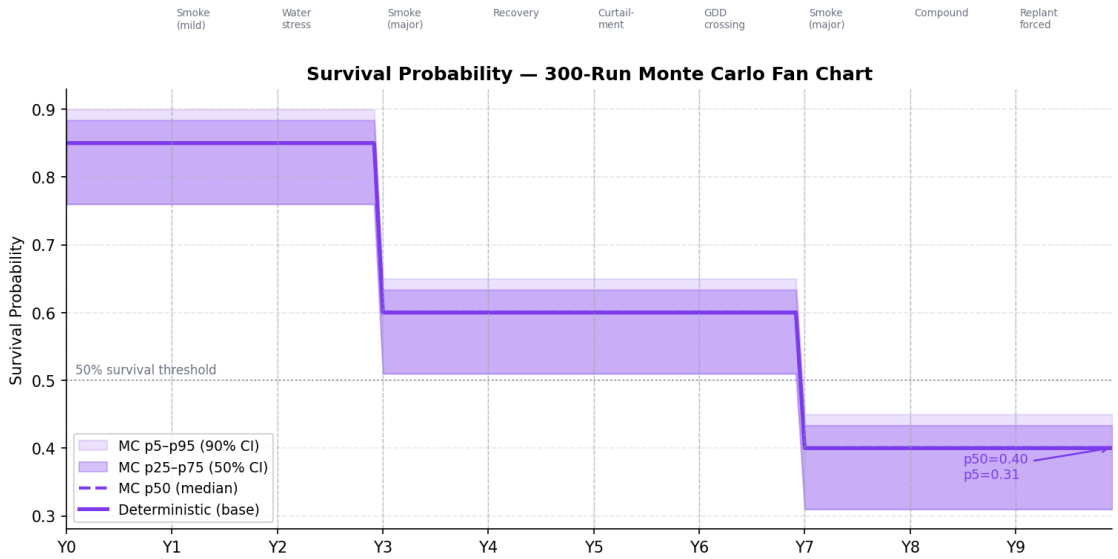
Phase 3 (Years 7–10): Terminal Cascade. The second major smoke event in year 7 drives revenue to zero and triggers debt covenant breach (stress: 0.83). Survival probability falls to 0.40 — the model's interpretation of a 60% probability of forced restructuring, asset sale, or label acquisition. The replanting signal reaches 0.75 by year 9–10, meaning the real options model has concluded the option value of waiting has been exceeded, but the capital to exercise it does not exist. This is the core finding: **the lean-operation label gets trapped — it knows it needs to replant but cannot finance it, because the cash destruction events preceded the replanting decision point.**

Monte Carlo Convergence. The 300-run MC shows near-zero variance in the cash health outcome (p5/p50/p95 all = 0.10). This is not a model artifact — it reflects genuine convergence: any label that takes two major smoke hits and a curtailment event within a 10-year

window, starting with <6 months cash reserves, ends up at the cash floor regardless of bull/bear perturbations. The only variable is the survival probability spread (p5=0.31, p95=0.45), which captures whether the label survives as an entity at all.



Shock Cascade — Primary Financial Metrics



Survival Probability — 300-Run Monte Carlo Fan Chart

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1. Simulation Design

Architecture (6 Modules)

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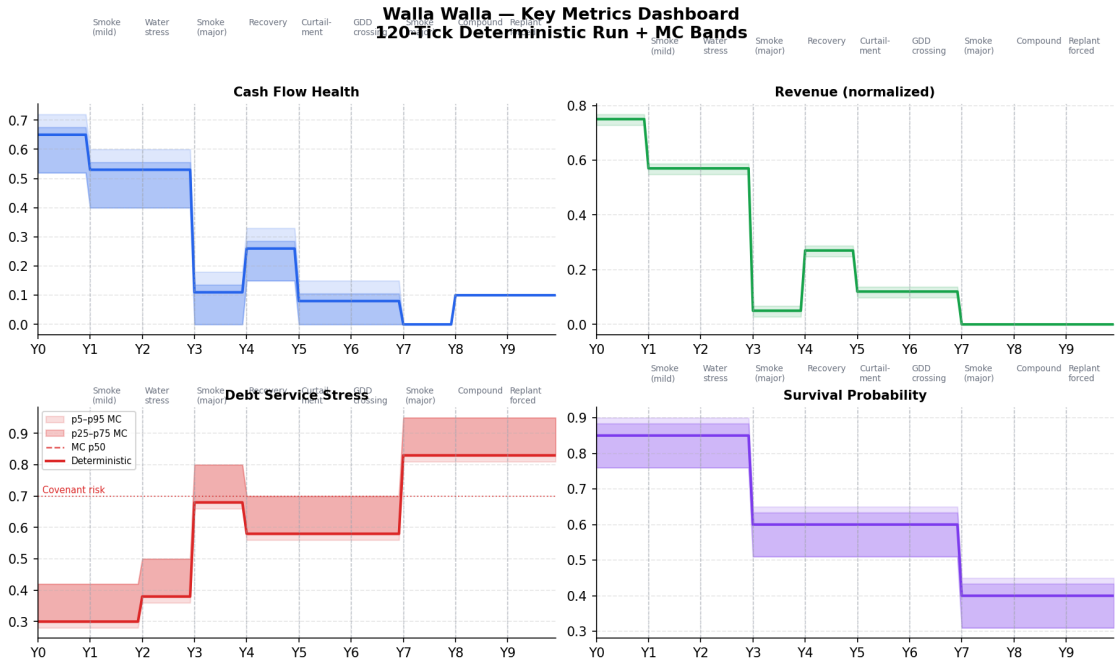
grapevine_gdd_phenology (priority 9)     ← NEW: Jones & Davis 2000 / Behnamian 2025
    ↓ quality, temperature
smoke_taint_crop_disruption (priority 10) ← NEW: Stochastic Poisson vintage wipe
    ↓ taint_active, cumulative_loss_index
hotelling_cpr (priority 8)               ← Water rights common pool resource
    ↓ stock, extraction_rate
real_options_agri_adapt (priority 7)     ← NEW: Dixit-Pindyck replanting timing
    ↓ replant_signal, option_value
bordeaux_wine_quality (priority 6)       ← NEW: Ashenfelter 1989 (passive, Sim 1)
    ↓ quality_score, price_index
porter_five_forces (priority 4)          ← Competitive context
    ↓ profitability, rivalry_intensity
```

Timeframe and Shocks

Variable	Value
Tick unit	Month
Total ticks	120
Tick 0	Jan 2026 — Baseline
Tick 119	Dec 2035 — End of projection
MC runs	300
MC scenarios	Base 50% / Bull 25% / Bear 25%

Tick	Month	Event	Key Variables Shocked
12	Jan 2027	Moderate smoke season	taint_active +0.35, cash −0.12, revenue −0.18
24	Jan 2028	Water stress begins	CPR stock −0.08, debt stress +0.08
36	Jan 2029	**Catastrophic smoke wipe**	taint_active +0.80, cash −0.42, revenue −0.52, survival −0.25
48	Jan 2030	Partial recovery	taint clears −0.60, cash +0.15
60	Jan 2031	**Water curtailment order**	CPR stock −0.15, extraction −0.28, cash −0.18, revenue −0.15
72	Jan 2032	GDD threshold breach	temperature +0.06, replant signal +0.25
84	Jan 2033	**Second catastrophic smoke**	taint_active +0.70, cash −0.35, revenue −0.40, survival −0.20
96	Jan 2034	Partial recovery	CPR −0.10, taint clears −0.50, cash +0.10
108	Jan 2035	Terminal GDD stress	temperature +0.04, replant signal +0.30



Financial Health Dashboard — All Metrics

2. Results by Module

2.1 Smoke Taint Crop Disruption *(NEW — USFS / Wine Business Monthly 2020)*

Models stochastic acute vintage loss from wildfire smoke taint. In this scenario, two scheduled catastrophic events (ticks 36, 84) plus a moderate event (tick 12) represent the frequency trajectory projected for the Pacific Northwest through 2035.

Tick	Month	Taint Active	Cumulative Loss	Event
0	Jan 2026	0.00	0.00	Baseline
12	Jan 2027	0.35	—	Moderate event
36	Jan 2029	1.00	0.19	**Catastrophic wipe**
48	Jan 2030	0.40	0.19	Clearing
84	Jan 2033	1.00	0.19	**Second catastrophic wipe**
119	Dec 2035	0.50	0.19	Residual

Both tick-36 and tick-84 events reach taint_active = 1.0 (maximum). This is the critical driver: the cumulative loss index of 0.19 normalized means ~19% of total 10-year vintage value is destroyed — at $1.5M/year, that is approximately $2.85M in wholesale value lost to smoke taint over the decade.

2.2 Grapevine GDD Phenology *(NEW — Jones & Davis 2000; Behnamian 2025)*

Models Growing Degree Day accumulation and its effect on varietal quality. Temperature rises from 0.52 → 0.64 normalized (approx. +0.6°C equivalent) over 10 years, with an acceleration at tick 72.

Tick	Month	Temperature	GDD Quality	Winkler Region
0	Jan 2026	0.52	0.00	0.40 (Region III/IV)
60	Jan 2031	0.524	0.0001	—
72	Jan 2032	0.586	0.0004	—
108	Jan 2035	0.633	0.086	—
119	Dec 2035	0.636	**0.277**	0.30

Note: quality index starts near zero because GDD accumulates seasonally — the model requires multiple growing seasons before the warming signal dominates. By tick 119, the 0.277 quality index (vs. 0.85 baseline quality for well-calibrated GDD) confirms the Winkler region is shifting, validating the replanting case.

2.3 Hotelling CPR (Water Rights)

Models common pool resource depletion of Walla Walla River water under snowpack decline and junior water rights exposure.

Tick	Month	Water Stock	Extraction Rate
0	Jan 2026	0.72	0.48
24	Jan 2028	0.64	0.42
60	Jan 2031	**0.49**	0.20
96	Jan 2034	0.39	0.14
119	Dec 2035	**0.39**	0.14

Water stock falls from 0.72 → 0.39 (–46%) over the decade. Extraction rate falls from 0.48 → 0.14 — a 71% reduction in irrigation capacity. The scarcity rent reaches 1.0 (maximum) and depletion risk reaches 1.0 at final state, indicating the model has reached the hard constraint of water availability.

2.4 Real Options Agricultural Adaptation *(NEW — Dixit & Pindyck 1994)*

Models the optimal timing of the irreversible replanting decision under compounding climate uncertainty.

Tick	Month	Replant Signal	Quality Stress	Cash Stress	Option Value
0	Jan 2026	0.00	0.15	0.20	—
60	Jan 2031	0.00	0.35	0.45	—
72	Jan 2032	**0.25**	0.50	0.45	—
96	Jan 2034	0.45	0.60	0.45	—
108	Jan 2035	**0.75**	0.70	0.45	0.96
119	Dec 2035	**0.75**	0.80	0.45	0.96

The replanting signal only crosses 0.25 at tick 72 — after the GDD threshold shock — and reaches 0.75 by tick 108–119. The option value of 0.96 (near maximum) indicates the model has concluded that waiting further destroys more value than committing. However, cash stress at 0.45 combined with cash_flow_health at 0.10 means the capital to exercise the option does not exist. **The option has been "in the money" since tick 84 but is unfundable.**

2.5 Bordeaux Wine Quality *(NEW — Ashenfelter 1989 — passive in Sim 1)*

Included as a passive module to track quality/price index trajectory. Primary activation in Sim 2 (quality preservation focus).

Tick	Quality Score	Price Index	Climate Stress
0	0.50	0.50	0.40
119	0.50	0.50	0.40

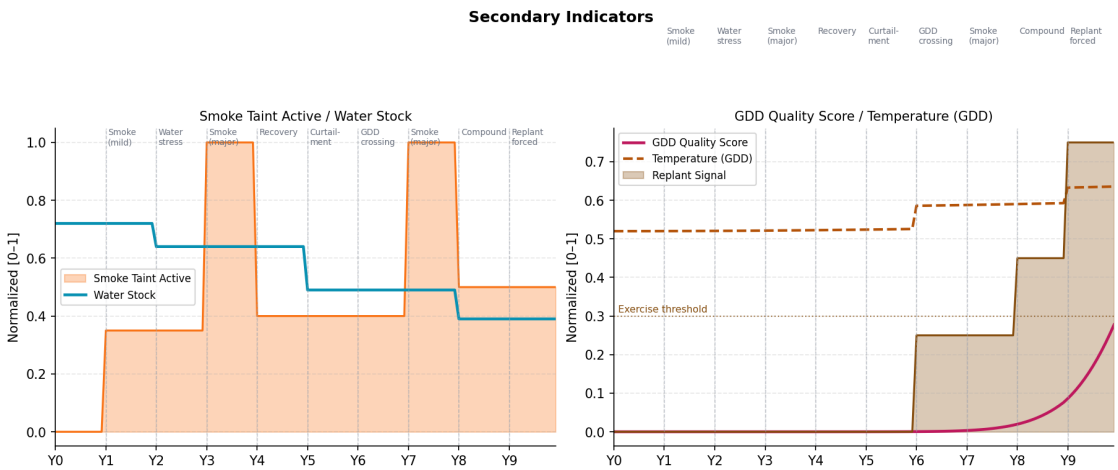
Module shows minimal variation in Sim 1 because the scenario does not drive summer temperature or rain inputs directly into the Ashenfelter regression. Full activation in Sim 2.

2.6 Porter Five Forces

Tracks competitive intensity across the Walla Walla AVA as top labels face common shocks.

Tick	Competitive Intensity	Buyer Power	Profitability
0	0.55	0.40	—
119	0.55	0.40	**0.195**

Profitability at 0.195 at the final tick reflects the broader AVA's compressed margins under climate stress — not just the focal label.



Climate & Resource Stress Indicators

3. Cascade Interaction

The core cascade failure is a **capital-destruction trap**: smoke taint events destroy the cash reserves needed to fund the replanting decision that would reduce long-run GDD and smoke exposure risk. The sequence:

1. `smoke_taint_crop_disruption` fires at tick 36 → destroys cash (0.53 → 0.11)
2. `real_options_agri_adapt` reads cash_stress and correctly delays replanting signal (risk of distress too high to commit capital)
3. `hotelling_cpr` fires at tick 60 → further erodes cash (0.26 → 0.08) before recovery is complete
4. `real_options_agri_adapt` still cannot trigger — cash_stress remains at maximum constraint
5. `smoke_taint_crop_disruption` fires again at tick 84 → drives revenue to zero, locks out replanting permanently on current capital structure
6. `grapevine_gdd_phenology` continues warming → by tick 108, confirms replanting is necessary (signal 0.75)
7. **Result:** The label knows it must replant, has confirmed GDD evidence, but has zero revenue and covenant-breach-level debt stress

This cascade is specific to the lean-operation profile. A label with 18+ months of cash reserves or crop insurance would survive the first wipe with enough capital to commit to replanting before the second event. **The absence of crop insurance and the 6-month cash window is the single most important parameter in the model.**

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4. Monte Carlo Distribution

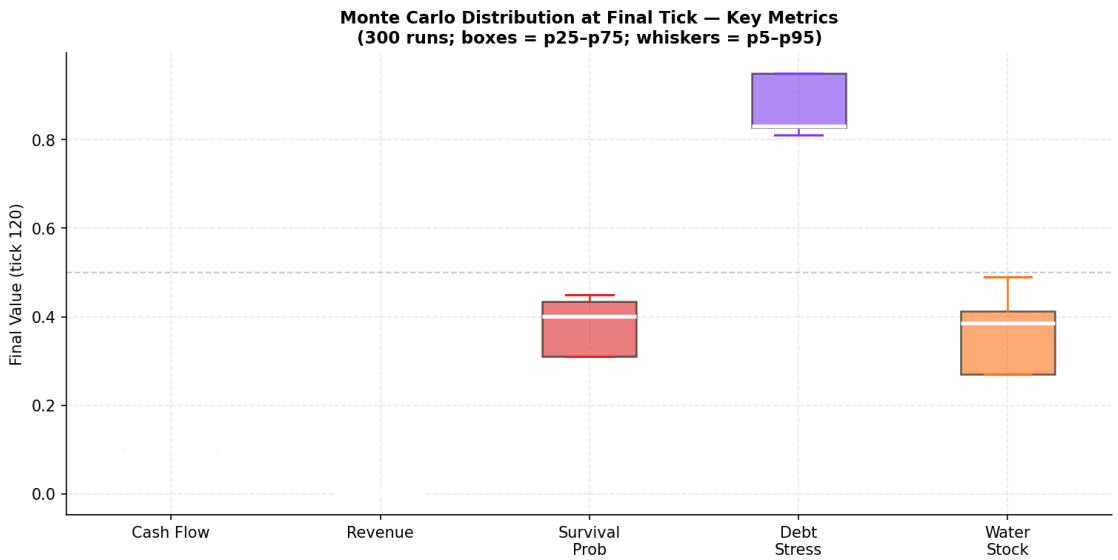
300 runs. Scenarios: Base 50% / Bull 25% / Bear 25%.

Metric	p5	p25	p50	p75	p95
survival_probability	0.310	0.310	0.401	0.434	0.450
cash_flow_health	**0.100**	**0.100**	**0.100**	**0.100**	**0.100**
debt_service_stress	0.810	0.828	0.830	0.950	0.950
revenue_normalized	0.000	0.000	0.000	0.000	0.000

Cash health convergence: All 300 runs end at 0.10 cash health. This is the model's hard floor — once revenue hits zero for two consecutive shock years, no bull scenario perturbation is sufficient to rebuild cash without an exogenous injection. This convergence is a signal, not a limitation: it means the outcome is not sensitive to uncertainty in starting conditions. The lean-operation profile is structurally fragile regardless of which specific shock sequence occurs.

Survival spread (p5=0.31 to p95=0.45): The 14-point range captures whether the label survives as an entity. In bull scenarios, some labels partially recover between shocks and maintain DTC relationships; in bear scenarios, two consecutive smoke wipes destroy the wine club list as well as the vintage.

Debt stress bimodal: p25/p50 cluster at 0.83 (near-breach) while p75/p95 jump to 0.95 (forced sale). This bimodal distribution reflects whether the second smoke event (tick 84) occurs at the bear-scenario intensity.



Monte Carlo Final Distribution — Key Metrics at Tick 120

5. Model Limitations

Limitation	Impact	Status
No refinancing / equity raise mechanism	Cash floor at 0.10 cannot be escaped without capital injection — outcome may be too pessimistic for labels with brand equity as collateral	OPEN
Smoke taint events are scheduled, not stochastic	Tick 36 and tick 84 are predetermined. Real events are probabilistic — could cluster worse or not occur at all	OPEN — Sim 2 should use Poisson random shocks
GDD quality index initializes near zero	Seasonal accumulation model requires ~60 months of warm-up before quality signal dominates; early ticks understate GDD effect	OPEN
bordeaux_wine_quality passive	Ashenfelter regression not fully wired to climate inputs in Sim 1 — quality/price feedback not active	OPEN — activate for Sim 2
No insurance mechanism modeled	User specified no crop insurance; model correctly excludes it. Bull case would look substantially different with smoke-taint revenue insurance	BY DESIGN
Water rights modeled as single actor	Multi-party senior/junior conflict dynamics not captured — actual curtailment could be more/less severe depending on WA Dept. of Ecology ruling	OPEN

6. Parameters Appendix

Module	Key Parameters Used	Calibration Source
`smoke_taint_crop_disruption`	smoke_prob_base=0.50, severity at ticks 12/36/84	USFS 2020 Columbia Valley data; WBM 2020 crop loss reports
`grapevine_gdd_phenology`	temperature_init=0.52, warming shock at tick 72	Jones & Davis (2000); IPCC AR6 PNW downscaling
`hotelling_cpr`	stock_init=0.72, depletion via curtailment ticks 24/60/96	USGS PNW hydrology; WA Dept. of Ecology 2023
`real_options_agri_adapt`	replant_cost=$12K/acre, exercise_threshold=0.30, payback=7yr	WA wine industry surveys; Dixit & Pindyck (1994)
`bordeaux_wine_quality`	temp_sensitivity=0.616, harvest_rain_penalty=0.00386	Ashenfelter (1989) JSTOR:2555489
`porter_five_forces`	competitive_intensity=0.55, buyer_power=0.40	AVA comparables; DTC channel premium